

COMPOSITE MATERIAL DATASHEETS CORE N636 HEXAGONAL HONEYCOMB

MATERIAL IDENTIFICATION

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Short informal name	N636-HEX
Commercial name	Hexcel HRH 36, Eurocomposites ECK, Schütz Cornmaster N636 CN1 or McGill
Specifications	AIMS11-01-004 (Grades B, C, F & G), Z-17.680/1/4/5, ABS5341A1/2/5/6
Description	Honeycomb Cores, N636 Polyamide Paper (meta and para-aramid), Phenolic Resin Impregnated, Hexagonal.
Version	1.3

REFERENCES

- ME0909581 Is. 3.0 "Design Values of N636 Hexagonal Honeycomb Cores" Airbus Technical Memorandum.
- RP0813804 Is. 1.0 "Method for the analysis of composite sandwich structures" Airbus Technical Report.
- TEA-A4-NT-210317 Is. B "A400M Cowlings. Bonded out-of-autoclave repairs. Test Results Analysis" Airbus DS Technical Report.
- TEPA-NT-2310002 Is. A "A400M Cowlings. Corrections in Design Values due to acceptable defects during manufacturing. Test Results Analysis" Airbus DS Technical Report.

GRADE B

PHYSICAL PROPERTIES

Cell size	4.8 mm
Density	32 kg/m ³
Max. Operational Limit (MOL)	135 °C

CTE (1/°C)	Range	-55..20°C	20..70°C	70..135°C
	α	-	-	-

ELASTIC MODULI (all manufacturers except McGill)

Property	Mean Values (MPa)					B-basis K_B	E-KDF			B-basis Values (MPa)				
	-55°C/Dry	RT/Dry	70°C/W	100°C/W	120°C/W		70°C/W	100°C/W	120°C/W	-55°C/Dry	RT/Dry	70°C/W	100°C/W	120°C/W
E_{33c}	34.0	34.0	32.0	29.0	18.0	0.74	0.941	0.852	0.528	25.1	25.1	23.6	21.4	13.3
G_{13}	76.3	76.3	71.0	65.0	40.3	0.74	0.930	0.851	0.528	56.4	56.4	52.5	48.1	29.8
G_{23}	38.0	38.0	36.0	34.0	20.1	0.74	0.947	0.894	0.528	28.1	28.1	26.6	25.1	14.8

STRENGTH (all manufacturers except McGill)

Property	Mean Values (MPa)					B-basis K_B	E-KDF			B-basis Values (MPa) (*)				
	-55°C/Dry	RT/Dry	70°C/W	100°C/W	120°C/W		70°C/W	100°C/W	120°C/W	-55°C/Dry	RT/Dry	70°C/W	100°C/W	120°C/W
F_{33c}	-	-	-	-	-	0.74	0.925	0.864	0.814	0.81	0.81	0.75	0.70	0.66
F_{13}	-	-	-	-	-	0.74	0.838	0.677	0.548	0.62	0.62	0.52	0.42	0.34
F_{23}	-	-	-	-	-	0.74	0.947	0.842	0.763	0.38	0.38	0.36	0.32	0.29

(*) These Design Values are only valid for core thickness below 12.7 mm. A thickness correction factor shall be applied as follows for core thickness above 12.7 mm: $F_{33c} \text{ (corrected)} = F_{33c} \cdot 1.29 \cdot t_c^{-0.1}$, $F_{13} \text{ (corrected)} = F_{13} \cdot 2.14 \cdot t_c^{-0.3}$
 $F_{23} \text{ (corrected)} = F_{23} \cdot 2.14 \cdot t_c^{-0.3}$; where t_c is the core thickness.

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STRENGTH (Hexcel & Eurocomposites only)

Property	Mean Values (MPa)				B-basis K_B	<i>E-KDF</i>		B-basis Values (MPa) (*)			
	-55°C/Dry	RT/Dry	70°C/W	100°C/W		70°C/W	100°C/W	-55°C/Dry	RT/Dry	70°C/W	100°C/W
F_{33c}	-	-	-	-	-	0.882	0.835	0.85	0.85	0.75	0.71
F_{13}	-	-	-	-	-	0.913	0.826	0.69	0.69	0.63	0.57
F_{23}	-	-	-	-	-	0.808	0.893	0.47	0.47	0.38	0.42

(*) These Design Values are only valid for core thickness below 12.7 mm. A thickness correction factor shall be applied as follows for core thickness above 12.7 mm: $F_{33c} \text{ (corrected)} = F_{33c} \cdot 1.29 \cdot t_c^{-0.1}$, $F_{13} \text{ (corrected)} = F_{13} \cdot 2.14 \cdot t_c^{-0.3}$, $F_{23} \text{ (corrected)} = F_{23} \cdot 2.14 \cdot t_c^{-0.3}$; where t_c is the core thickness.

OTHER PROPERTIES

M-KDF's are provided in TEPA-NT-2310002 Is. A for typical quality deviations and dedicated wrinkling coefficients are provided in TEA-A4-NT-210317 Is. B if combined with Hexcel T300/F593 material.

COMPOSITE MATERIAL DATASHEETS CORE N636 HEXAGONAL HONEYCOMB

GRADE C

PHYSICAL PROPERTIES

Cell size	4.8 mm
Density	37 kg/m ³
Max. Operational Limit (MOL)	135 °C

CTE (1/°C)	Range	-55..20°C	20..70°C	70..135°C
	α	-	-	-

ELASTIC MODULI (all manufacturers except McGill)

Property	Mean Values (MPa)				B-basis K_B	<i>E-KDF</i>		B-basis Values (MPa)			
	-55°C/Dry	RT/Dry	70°C/W	100°C/W		70°C/W	100°C/W	-55°C/Dry	RT/Dry	70°C/W	100°C/W
E_{33c}	43.0	43.0	36.0	32.0	0.74	0.837	0.744	31.8	31.8	26.6	23.6
G_{13}	86.0	86.0	73.0	65.0	0.74	0.848	0.755	63.6	63.6	54.0	48.1
G_{23}	43.0	43.0	38.0	35.8	0.74	0.883	0.832	31.8	31.8	28.1	26.4

STRENGTH (all manufacturers except McGill)

Property	Mean Values (MPa)				B-basis K_B	<i>E-KDF</i>		B-basis Values (MPa) (*)			
	-55°C/Dry	RT/Dry	70°C/W	100°C/W		70°C/W	100°C/W	-55°C/Dry	RT/Dry	70°C/W	100°C/W
F_{33c}	-	-	-	-	-	0.612	0.564	1.24	1.24	0.76	0.70
F_{13}	-	-	-	-	-	0.866	0.693	0.75	0.75	0.65	0.52
F_{23}	-	-	-	-	-	0.875	0.770	0.48	0.48	0.42	0.37

(*) These Design Values are only valid for core thickness below 12.7 mm. A thickness correction factor shall be applied as follows for core thickness above 12.7 mm: $F_{33c}(\text{corrected}) = F_{33c} \cdot 1.29 \cdot t_c^{-0.1}$, $F_{13}(\text{corrected}) = F_{13} \cdot 2.14 \cdot t_c^{-0.3}$, $F_{23}(\text{corrected}) = F_{23} \cdot 2.14 \cdot t_c^{-0.3}$; where t_c is the core thickness.

OTHER PROPERTIES

None.

COMPOSITE MATERIAL DATASHEETS CORE N636 HEXAGONAL HONEYCOMB

GRADE F

PHYSICAL PROPERTIES

Cell size	3.2 mm
Density	48 kg/m ³
Max. Operational Limit (MOL)	135 °C

CTE (1/°C)	Range	-55..20°C	20..70°C	70..135°C
	α	-	-	-

ELASTIC MODULI (all manufacturers)

Property	Mean Values (MPa)					B-basis K_B	<i>E-KDF</i>			B-basis Values (MPa)				
	-55°C/Dry	RT/Dry	70°C/W	100°C/W	120°C/W		70°C/W	100°C/W	120°C/W	-55°C/Dry	RT/Dry	70°C/W	100°C/W	120°C/W
E_{33c}	63.0	63.0	59.0	53.0	28.4 (*)	0.74	0.936	0.841	0.45 (*)	46.6	46.6	43.6	39.2	21.0
G_{13}	119.5	119.5	112.0	102.0	53.8 (*)	0.74	0.937	0.853	0.45 (*)	88.4	88.4	82.8	75.4	39.8
G_{23}	57.8	57.8	56.0	53.0	26.0 (*)	0.74	0.968	0.916	0.45 (*)	42.7	42.7	41.4	39.2	19.2

(*) Values conservatively derived by comparison with Grade B equivalents.

STRENGTH (all manufacturers)

Property	Mean Values (MPa)					B-basis K_B	<i>E-KDF</i>			B-basis Values (MPa) (**)				
	-55°C/Dry	RT/Dry	70°C/W	100°C/W	120°C/W		70°C/W	100°C/W	120°C/W	-55°C/Dry	RT/Dry	70°C/W	100°C/W	120°C/W
F_{33c}	-	-	-	-	-	-	0.863	0.801	0.70 (*)	2.27	2.27	1.96	1.82	1.58
F_{13}	-	-	-	-	-	-	0.833	0.666	0.45 (*)	1.26	1.26	1.05	0.84	0.56
F_{23}	-	-	-	-	-	-	0.897	0.794	0.65 (*)	0.68	0.68	0.61	0.54	0.44

(*) Values conservatively derived by comparison with Grade B equivalents.

(**) These Design Values are only valid for core thickness below 12.7 mm. A thickness correction factor shall be applied as follows for core thickness above 12.7 mm: $F_{33c}(\text{corrected}) = F_{33c} \cdot 1.29 \cdot t_c^{-0.1}$, $F_{13}(\text{corrected}) = F_{13} \cdot 2.14 \cdot t_c^{-0.3}$, $F_{23}(\text{corrected}) = F_{23} \cdot 2.14 \cdot t_c^{-0.3}$; where t_c is the core thickness.

OTHER PROPERTIES

None.

COMPOSITE MATERIAL DATASHEETS CORE N636 HEXAGONAL HONEYCOMB

GRADE G

PHYSICAL PROPERTIES

Cell size	4.8 mm
Density	48 kg/m ³
Max. Operational Limit (MOL)	135 °C

CTE (1/°C)	Range	-55..20°C	20..70°C	70..135°C
	α	-	-	-

ELASTIC MODULI (all manufacturers except McGill)

Property	Mean Values (MPa)				B-basis K_B	<i>E-KDF</i>		B-basis Values (MPa)			
	-55°C/Dry	RT/Dry	70°C/W	100°C/W		70°C/W	100°C/W	-55°C/Dry	RT/Dry	70°C/W	100°C/W
E_{33c}	63.0	63.0	56.0	51.0	0.74	0.888	0.809	46.6	46.6	41.4	37.7
G_{13}	121.0	121.0	111.0	98.0	0.74	0.917	0.809	89.5	89.5	82.1	72.5
G_{23}	61.0	61.0	55.0	52.0	0.74	0.901	0.852	45.1	45.1	40.7	38.4

STRENGTH (all manufacturers except McGill)

Property	Mean Values (MPa)				B-basis K_B	<i>E-KDF</i>		B-basis Values (MPa) (*)			
	-55°C/Dry	RT/Dry	70°C/W	100°C/W		70°C/W	100°C/W	-55°C/Dry	RT/Dry	70°C/W	100°C/W
F_{33c}	-	-	-	-	-	0.816	0.777	2.02	2.02	1.65	1.57
F_{13}	-	-	-	-	-	0.950	0.800	1.20	1.20	1.14	0.96
F_{23}	-	-	-	-	-	0.900	0.866	0.60	0.60	0.54	0.52

(*) These Design Values are only valid for core thickness below 12.7 mm. A thickness correction factor shall be applied as follows for core thickness above 12.7 mm: $F_{33c} \text{ (corrected)} = F_{33c} \cdot 1.29 \cdot t_c^{-0.1}$, $F_{13} \text{ (corrected)} = F_{13} \cdot 2.14 \cdot t_c^{-0.3}$
 $F_{23} \text{ (corrected)} = F_{23} \cdot 2.14 \cdot t_c^{-0.3}$; where t_c is the core thickness.

OTHER PROPERTIES

None.